

Aditya Kaushikbhai Rathod

B.Tech. in Electrical Engineering | Indian Institute of Technology, Jodhpur

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EDUCATION

Indian Institute of Technology, Jodhpur

B.Tech. in Electrical Engineering | CGPA: 7.38/10

Relevant Coursework: Data Structures & Algorithms, Digital Design, Pattern Recognition & Machine Learning, Probability, Statistics & Stochastic Processes, Mathematics I & II, Introduction to Computer Science, Foundation of Quantum Information

PROFESSIONAL SUMMARY

- Software engineering student and open-source contributor focused on backend systems, Rust/Python development, and AI developer tooling; built CORTEX, a **180+ endpoint FastAPI** backend across **10 domain routers**, validated with **1,900+ automated tests**.
- Built ASTERA, a Rust code-intelligence engine that parses **14 programming languages** with Tree-sitter for repository-scale static analysis, semantic search, and impact reasoning; validated with **153 automated tests** and **92 Criterion benchmarks**.
- Evaluating **30+ CNN-Attention architectures** in PyTorch through controlled ablations on the imbalanced PlantVillage and Mendeley Potato Leaf Disease datasets.

TECHNICAL SKILLS

Languages: Python, C, C++, Rust, TypeScript, JavaScript, SQL

Systems: Data Structures, Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Socket Programming, TCP/IP, Multithreading, Concurrency, Async Programming, Memory Management

Backend & Infrastructure: FastAPI, Axum, REST APIs, WebSockets, Microservices, Pydantic, SQLAlchemy, Alembic, Redis, Linux, Docker, Docker Compose, GitHub Actions, CI/CD, API Design

Databases & Search: PostgreSQL, SQLite, Qdrant, pgvector, SQLite FTS5, Vector Search, Semantic Search, Hybrid Search, BM25, HNSW

AI & Developer Tools: PyTorch, OpenCV, YOLO, Ollama, RAG, Embeddings, AI Agents, Tool Calling, Model Evaluation, Ablation Studies, Tree-sitter, Static Analysis, Program Analysis, Code Property Graphs

Engineering: Git, pytest, Serde, Clap, Notify, Tracing, Unit Testing, Integration Testing, Debugging, Code Review, Authentication, Rate Limiting, Cryptography (AES-256, SHA-256), Criterion Benchmarking

EXPERIENCE

Undergraduate Researcher – Computer Vision, IIT Jodhpur

| May 2026 – Present | [GitHub](#)

Robust Classification for Imbalanced Natural Image Datasets

PyTorch, Computer Vision, Reproducible ML Evaluation

- Reproduced the PLDNet baseline in PyTorch with fixed dataset splits, training settings, and evaluation scripts, so every comparison is like-for-like.
- Evaluating **30+ model variants** using ablations and feature-fusion changes to compare robustness under class imbalance and real-world image variation.
- Analyzing Mendeley Potato Leaf Disease and PlantVillage results by class-imbalance behavior, not headline accuracy alone.

Open Source Contributor – plink, Devlup Labs IIT Jodhpur

| May 2025 – Nov. 2025 | [GitHub](#)

Secure Peer-to-Peer File Transfer across Heterogeneous Networks

Python, TCP/IP, Sockets, Multithreading, AES-256, NAT Traversal, Git

- Contributed TCP socket paths for AES-256 encrypted file chunks, metadata exchange, and multithreaded transfer across heterogeneous networks; commits merged and plink is live and usable.
- Completed **16 assigned tasks** across cryptography, networking research, file metadata, chunk handling, parsing refactors, port scanning, UDP hole-punching prototypes, direct-connection management, network-type detection, and documentation.

PROJECTS

• ASTERA

| Mar. 2026 – Present | [GitHub](#)

Multi-Language Static Analysis & Code Intelligence Engine

Rust, Tree-sitter, SQLite FTS5, Axum, Tokio, Rayon, REST APIs, Program Analysis

- Built a **12-crate Rust pipeline** (file discovery, Tree-sitter parsing, symbol extraction, graph edges, SQLite FTS5 indexing, REST APIs, exports) that parses **14 languages** into a queryable **Code Property Graph**.
- Enabled symbol lookup, dependency traversal, impact analysis, broken-reference detection, and semantic search.
- Optimized hot paths to **33 ns** language classification, **16.16 µs** SQLite FTS5 symbol search, **3–24 µs** impact traversals, and **229 ms** metrics over 10,000 nodes.
- Validated parser, storage, REST API, search, and graph traversal paths with **153 automated tests** and **92 Criterion benchmarks**.

• CORTEX

| Jan. 2026 – Present | [GitHub](#)

Local-First Machine Intelligence Layer

Python, FastAPI, Next.js, PostgreSQL, Redis, Qdrant, JWT, Agent Systems

- Designed **hybrid retrieval** combining semantic, episodic, and working memory, vector search, and knowledge-graph recall over repositories, documents, conversations, and project history.
- Implemented asynchronous services for repository analysis, file tracking, GitHub sync, memory search/decay, model catalogs, providers, and recommendations in a **180+ endpoint FastAPI backend**.
- Implemented the **agent run loop** covering tool execution, output verification, stall detection, context compaction, token accounting, and security policy checks.
- Validated agents, REST APIs, memory, privacy, services, and infrastructure with **1,900+ automated tests** across **10 domain routers**.

• Parallel Blockchain Simulator

| Nov. 2025 – Dec. 2025 | [GitHub](#)

Concurrent Blockchain Simulation with Thread-Safe Validation

Python, Multithreading, SHA-256, Queues, Hash Maps, Merkle Trees, Mutex Locks

- Coordinated **7 blockchain threads** (3 miners, 3 validators, 1 transaction generator) with mutex locks and events around shared chain state.
- Modeled balances, linked-list blocks, queued transactions, hash-map mempool, Merkle verification, and SHA-256 Proof-of-Work (difficulty 3).
- Stress-tested under randomized loads: **131–200 transactions** and **20–59 mined blocks** per run, chain never corrupted.

LEADERSHIP & CAMPUS INVOLVEMENT

Secretary, Designers – The Design Club, IIT Jodhpur

Aug. 2025 – Apr. 2026

- Guided **50+ students**, ran design workshops, and secured **7th rank** at Inter IIT Cultural Meet 8.0.

Student Guide, Student Wellbeing Committee (SWC), IIT Jodhpur

Aug. 2025 – Apr. 2026

- Supported **12 first-year students** through academic transition and campus onboarding.

Core Member, Inside – The Game Development Society, IIT Jodhpur

Aug. 2025 – Apr. 2026

- Coached **5 first-year students** on an Inside-backed game-building project.

Assistant Head, Prometeo'26 – IIT Jodhpur

Oct. 2025 – Jan. 2026

- Led **30+ volunteers** and managed digital design for Prometeo'26.